

Blood Flow Titration

Imagine that you are initiating a patient on extracorporeal membrane oxygenation (ECMO). The cannulas are in excellent position and the circuit is connected and ready to go. You release the clamp and dark blood flows out smoothly.

“Time on ECMO, 7:38 a.m.” you say.

Within a few seconds, bright red blood is infusing into the return cannula. You give a reassuring glance at the console – circuit pressures look good, flows are steady, everything is running well. Next, your eyes move to the monitor. The oxygen saturation begins to slowly improve, blood pressure is coming up, and the heart rate comes down.

As the cannulas are being secured and dressed, someone asks, “what settings should we start?” Now comes the real challenge...

We will now get into the nuts and bolts of the daily management of ECMO. We have said again and again that ECMO really only involves the manipulation of a few variables, namely, sweep gas and blood flow. Now that you are armed with the physiologic principles at play, you can start to develop some management principles for how to titrate flow and sweep for your patients on ECMO.

THE INS AND OUTS OF FLOW TITRATION

Blood flow determines the amount of oxygen delivered by increasing the relative amount of oxygenated blood from the circuit relative to the overall output of the native heart and lungs. The higher the flow, the more oxygenated blood is being delivered to the body. Flow titration requires identification of the ideal amount of blood flow through the ECMO circuit required to meet the desired physiologic end points.

HOW MUCH FLOW IS ENOUGH? HOW MUCH IS TOO MUCH?

The ideal blood flow can be hard to define. While the individual physiologic needs and limits of the patient are the primary consideration, other considerations may come down to the style of the individual practitioners/programs. Needless to say, let's explore some principles that can be applied when trying to reach this flow.

As you increase flow, you will likely run into the following limits:

1. Minimum flow needed to maintain the pump
2. Achievement of a physiologic end point
3. You have reached a physiologic limit
4. You have reached a limit of the pump

Once you define these limits for each individual patient, you can better understand the ideal amount of flow to apply. Just because you *can* flow more doesn't mean you *should* flow more.

Limit #1: Minimum Flow Needed to Maintain the Pump

When you are first initiating ECMO, before the last clamp is released, there will be a minimum revolutions per minute (RPM) that is maintained, usually around 1500 RPM. Why is this? Why not just start at 0 and work your way up?

Remember when we were talking about the physiology of the pump, that forward flow is generated by the pressure differential from the pump. This pressure differential has to be enough to overcome the afterload of the pump, which comes in the form of the resistance of the circuit and either the mean arterial pressure (MAP) (in VA ECMO) or the central venous pressure (CVP) (in VV ECMO) (Fig. 17.1).

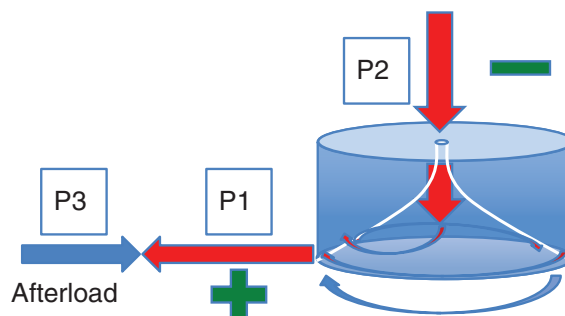


FIG. 17.1 The effect of afterload on pump function

At very low flows, the afterload may overtake the pressure generated by the pump at which you can have either no flow or retrograde flow. Additionally, the lower the flow, the greater the risk of stasis of blood and thrombosis in both the pump and in the oxygenator.

Put together, there is a minimum amount of blood flow that is maintained, usually no less than around 1.5 to 2 L in adults. Maintaining less flow may require a connection in the circuit separating the flow generated from the pump from the flow delivered to the patient (referred to as a bridge).

Limit #2: Achievement of a Physiologic Endpoint

Good critical care involves identifying a deficiency, applying an intervention, assessing the effect, and adjusting course. It is one thing to place a patient on fluids. It is much more effective to notice that a patient's labs, examination findings, and urine output are likely consistent with hypovolemia, give a bolus of fluid, and monitor the patient clinically to observe the clinical effect.

The same is true for flow, which is a form of support that can be dosed just like fluids, pressors, or mechanical ventilation. We spent a good deal of time in the ECMO Physiology section discussing and unpacking the physiologic rationale for ECMO, all of which are directly related to flow. Let's review:

Physiologic rationale for ECMO

VV ECMO:

- Clear CO₂
- Mitigate shunt physiology
- Optimize right heart function

VA ECMO:

- Optimize perfusion
- Support MAP
- Perfuse visceral organs
- Support right heart
- Provide oxygenation

VV ECMO: Recall that VV ECMO improves oxygen by mitigating shunt physiology. In respiratory failure, as shunt increases, more blood passes from the right side of the heart to the left side without participating in gas exchange. As ECMO blood flow increases, it takes on a higher percentage of this total blood flow that is crossing the pulmonary circulation, and oxygenation improves. In a similar way, while sweep gas flow is the primary determinant of CO_2 clearance, recall from [Chapter 11](#) that the higher the blood flow, the more CO_2 is cleared for every increase in sweep gas flow.

What does this mean for titration? The main take-home is that flow is related to DO_2 . If there is evidence of inadequate DO_2 (low venous oxygen saturation, tachycardia, hypoxia), then this is where we have to evaluate for higher flow. The decision to consider at this point is increasing flow, tolerating lower DO_2 , or giving blood.

VA ECMO: In [Chapter 13](#), we talked about the vicious cycle of cardiogenic shock and how VA ECMO flow can help to mitigate these factors. Now is the time to titrate these effects. MAP is low? Lactate is not clearing fast enough? Urine output is borderline? CVP and PA pressures are increasing showing evidence that the right heart is straining? These are the clinical indicators that more blood flow may be needed.

Remember, flow is like a pressor or any other drug that must be titrated. Notice a deficiency, titrate the flow up or down, and then observe if this change achieved the target effect. Now we are getting somewhere!

Limit #3: You Have Reached a Physiologic/Anatomic Limit

At a certain point, flow starts to adversely impact the physiology of the patient. This can be subtle or very obvious and can overlap with the beneficial effects of increased flow as described earlier. As such, it may be difficult to compile a comprehensive list, but for the meantime, let's explore some adverse physiologic implications of too much flow.

Excessive negative venous pressure: Remember that to generate flow, the centrifugal pump generates a negative venous pressure, sucking blood out of the venous system through the pump. As flow increases this negative pressure grows, pulling the vena cava and the venous structures towards the cannula ([Fig. 17.2](#)).

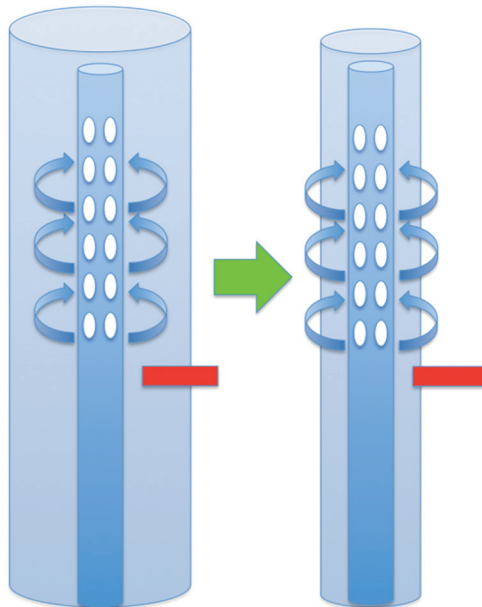


FIG. 17.2 Excessive negative pressure leading to collapse of the inferior vena cava

The effect of this can range anywhere from some mild kicking of the lines to dropping the flows for the circuit.

Biotrauma: Biotrauma refers to the inflammatory effect of support to tissues on a microscopic level. It is often described for mechanical ventilation, quantifying the damaging effects of mechanical ventilation that may be not accounted for by barotrauma and volutrauma effects alone. It is possible that higher blood flow through an artificial membrane oxygenator may lead to biotrauma as well, which can increase with increasing blood flow.

Recirculation (VV ECMO): This is a concept that we are now very familiar with but one that should be put in the context of flow titration. As flow increases on VV ECMO, there is more negative pressure generated by the drainage cannula, meaning there is a higher likelihood of pulling blood that is returning from the circuit back into the drainage cannula, as illustrated in Fig. 17.3. This has the dual effect of decreasing the effective ECMO blood flow and decreasing the efficiency of the membrane, by raising the saturation of blood entering the membrane.

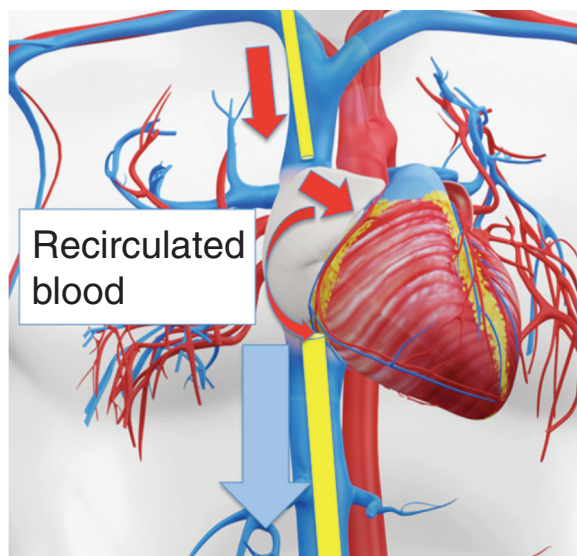


FIG. 17.3 Recirculation while on VV ECMO (Modified from [SciePro/Shutterstock.com](https://www.shutterstock.com).)

What does this mean for flow titration? It means that as we increase blood flow, we need to look for evidence of recirculation, such as higher saturation of pre-oxygenator blood, a narrowing gap between pre-oxygenator O_2 saturation and patient O_2 saturation, and narrowing gap between pre-oxygenator CO_2 and post-oxygenator CO_2 levels.

Right ventricular and pulmonary vascular effects (VV ECMO): While on VV ECMO, the net flow of blood to the right atrium should be even – any blood that is returned to the heart is taken away from the venous circulation. However, there remains the possibility that the flow direction and velocity adversely affects the right ventricle and the pulmonary circulation. The overall adverse impact of this effect is yet to be defined.

Retrograde flow and the left heart (VA ECMO): We described the adverse effects of retrograde blood flow in Chapter 14. The higher the retrograde flow that is applied, the greater the afterload that the left heart must work against. The heart's ability to accommodate this is largely influenced by the overall condition of the heart and the etiology causing the shock. This can

manifest as decreased pulsatility, higher probability of ventricular/valvular thrombus, increasing workload and worsening ischemia, and pulmonary edema with worsening oxygenation.

Some of these effects can be readily observed and quantified at the bedside, while others may be less readily apparent or even theoretical. The effects and limits of increasing blood flow can exist as a spectrum as illustrated in Fig. 17.4.

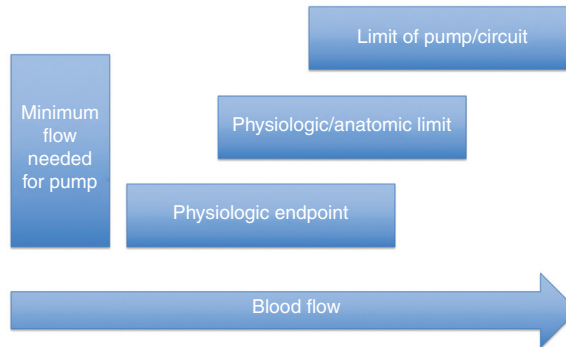


FIG. 17.4 Effects and limits of increasing blood flow

Does this mean that increasing blood flow is wrong at any given time? Not necessarily. What it does mean is that you should quantify the effect that you are aiming for and titrate flow to that effect. It is a subtle distinction, but one that will pay dividend if applied for your patients.

Limit #4: You Have Reached a Limit of the ECMO Pump/Circuit

Many of these limits have been explored in Chapter 10. To review:

Limit of the pump: The centrifugal pump generates forward flow by generating negative pressure on the drainage side and positive pressure on the return side. The greater the difference in these pressures, the greater the resulting flow will be. However, this is dependent on the speed of the pump – all pumps have a maximum amount of RPMs that they are able to generate. This is relevant to your patient because the higher the RPMs, the higher the probability that the pump speeds will be damaging to the blood that is crossing the pump.

There is no absolute number of RPMs that does or does not cause hemolysis, and this may change based on the configuration of the pump head/motor. Rather than aim for a specific number of RPMs, attempt to keep RPMs as low as possible to achieve the target flow (and more importantly target physiologic endpoint) and observe for evidence of hemolysis when RPMs are increased. If RPMs are being increased with a minimal increase in flow, this may be an indication that you are reaching the limits of the pump to increase flow.

Limit of the membrane: We introduced the concept of the rated flow of the membrane oxygenator – the flow at which no more oxygen transfer can occur. This is a standard value that is usually set by the manufacturer. Recall that as blood flow increases, the oxygen content transferred from the membrane increases, but at a decreasing rate. As flow increases, observe for evidence of diminishing returns by observing the oxygen levels of the post-oxygenator blood gas (Fig. 17.5).

Limit of the circuit/cannulas: The primary way the circuit/cannula limits flow is through resistance to flow, as described in our equation:

$$\text{Flow} = \frac{\pi \times \Delta P \times r^4}{8 \times \text{viscosity} \times \text{length}}$$

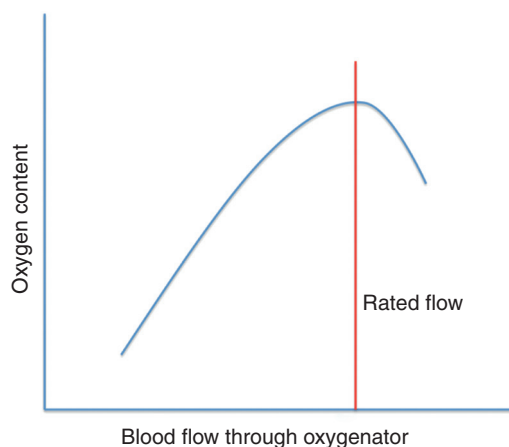


FIG. 17.5. Oxygenation capability of the oxygenator relative to blood flow

The narrower the cannulas and the longer the tubing, the more there will be resistance to flow. This will manifest as higher post-pump pressures and can eventually damage the circuit/oxygenator.

SEPARATING OUT PRELOAD FROM AFTERLOAD LIMITATIONS OF THE PUMP/CIRCUIT

We previously described the centrifugal pump as preload dependent and afterload sensitive. This means that preload is necessary for the pump to function and a lack of preload will rapidly drop the flow that the pump is able to generate. Afterload, on the other hand, will impact the overall flow of blood, dropping flow in proportion to the afterload that it must pump against.

Preload to the pump is determined by the intravascular volume of the venous system, size and compliance of the vena cava, pump speed, cannula size, and cannula position. The impact of cannula size and position is paramount, since it is often the variable that can be anticipated at selection. Larger diameter, multistage cannulas allow for more drainage of blood for any given speed of the pump and can allow for preload to the pump without excessive negative venous pressures. Cannula position can allow for adequate preload – a femoral drainage cannula placed in the intrahepatic inferior vena cava (IVC) can allow for drainage of blood with less vena caval collapse since the liver helps to stent the vena cava open.

Afterload to the pump, you will recall, is determined by resistance of the circuit/oxygenator/return cannula as well as the pressure of the system that blood is being returned to. For VV ECMO, this will be approximated by the CVP while for VA ECMO, this will be approximated by the MAP.

As a general principle, limitation to flow on VV ECMO will be on the negative side and is often due to the size of the drainage cannula while the limitation to flow on VA ECMO is more likely to be on the afterload side due to the return cannula.

This is not always the case. For example, there can be a lack of preload in VA ECMO, especially in the setting of hypovolemia/bleeding and VV ECMO can have excessive afterload, due to kinking of the lines, oxygenator clots, or coughing/clamping down. However, using this principle can

allow for a system of what to evaluate first, when you encounter a drop in flow in a patient on VV versus VA ECMO.

Features to suggest a limitation of preload include excessive negative venous pressures, flow jumping up and down, and flow drops that improve with winding down the RPMs administering fluid. By contrast, flow that drops and stays down and flow that does not improve with decreasing RPMs/administering fluid may be more likely to be associated with afterload limitations to flow.

What This Means for the Patient on a Day-To-Day Basis

As you can see, there are many variables that go into what flow you are running on a day-to-day basis. This can get complex very quickly and can lead to uncertainty in the best way to support the patient.

We will apply our same critical care management principles to flow – identify a deficiency, apply an intervention, assess the effect, and adjust course. When it comes to ECMO blood flow, try to identify what goals you are trying to accomplish with any flow, assess if those goals are being met, and what limits are holding you back.

For the most part, if the same physiologic endpoint can be achieved with a lower flow, go with the lower flow. Just because you can flow higher doesn't necessarily mean that you should flow higher.

This principle will serve well to limit any damaging effects of high blood flow and can potentially limit biotrauma. Additionally, it will allow for the natural progression to weaning, as detailed later.

WEANING ECMO BLOOD FLOW

As heart and lung recovery progresses, the ability of the body to adequately deliver oxygen improves to the point that ECMO may no longer be necessary, allowing for ECMO to be weaned off. This can be one of the most fulfilling parts of the care of any patient on ECMO. As the physiology of VV and VA ECMO is very different, we will discuss the strategies and rationale for weaning each separately.

Blood Flow Weaning and VV ECMO

As the lungs begin to improve, there is more participation of the alveolar unit in gas exchange. This leads to more V:Q matching and a lower shunt fraction. Now all of a sudden, blood that is pumped out of the right heart across the pulmonary circulation is not completely shunted across.

The result is the reverse of the process that we uncovered when we discussed respiratory failure – less intrapulmonary shunt means less blood flow from the ECMO circuit that is needed to capture the cardiac output. This will mean that you will be able to achieve physiologic endpoints of improved oxygen delivery (limit #2 above) without running into a limit from the ECMO pump or an anatomic/physiologic limit (limit #3 and #4 above).

However, at a certain point, you begin to reach the limit of what the pump needs to run. For some pumps this will be 3 L, for some 2.5 L or even lower. Regardless, there will be a point when lowering blood flow will not be sufficient to determine whether the patient is able to wean from VV ECMO. At this point, you must rely on sweep weaning (see [Chapter 18](#)).

Blood Flow Weaning and VA ECMO

Weaning ECMO blood flow is the primary mechanism of weaning for VA ECMO. As opposed to VV ECMO, which still relies on the ability of the right/left heart to provide cardiac output for oxygen delivery, overall perfusion is the sum of the native cardiac output and the flow from the machine, as follows:

$$CO_{ECMO} = \text{Native CO} + \text{ECMO blood flow}$$

Therefore, to truly understand the ability of the native heart and lungs, flow must be weaned down or even off briefly. The actual blood flow that can be tolerated for weaning will vary depending on the risk of clotting, the overall status of the patient, and the individual protocols of the institution. A patient with complete cardiac recovery and a robust MAP will likely need a much shorter weaning trial than a patient with borderline cardiac function.

Variables that can be accounted for while assessing the ability to wean include the following:

- **Vital signs:** MAP, heart rate
- **PA catheter pressures:** PA systolic, PA diastolic, CVP pulmonary capillary wedge pressure
- **Echocardiogram findings:** left ventricular function, right ventricular function, aortic valve opening, etc.
- **Clinical evidence of perfusion:** urine output, mental status, capillary refill
- **Laboratory findings:** venous oxygen saturation, lactic acid

The lower the flows (less than 2 L/min, for example) and the longer the amount of time needed for the weaning trial (hours v. minutes), the greater the risk of thrombus in the cannula, oxygenator, or circuit. Higher anticoagulation targets may be needed.

Why Can You Wean Flow to Assess for Weaning VA ECMO But Not for VV ECMO?

Let's summarize the difference in weaning blood flows in VV versus VA ECMO:

Weaning and VV/VV ECMO

VV ECMO: Flows may be weaned to a point where physiologic endpoints are met but not to assess for separation from support. To assess for removal, wean **sweep**

VA ECMO: Flow must be weaned to assess for ability to remove ECMO

The reason VV ECMO flows cannot be weaned to assess for decannulation is because the time period for weaning is often on the order of hours, which could be poorly tolerated with increased risk of thrombus and pump stoppage. Additionally, native CO_2 clearance is often slow to normalize in respiratory failure as compared to cardiac failure, meaning that sweep weaning will be a truer test of ability to wean rather than blood flow wean. We will go into sweep gas titration/weaning in more detail in the next chapter.

In VA ECMO, the contribution of ECMO blood flow to the overall perfusion is the main indication for support. If clinical endpoints can be met with low to no flow, then there is a high likelihood of successfully being able to separate from support and decannulate.

Now that we have a much better sense of how to titrate blood flow, we will now move into principles for titration of sweep gas flow.

SUGGESTED READING

- Shekar, K., Buscher, H., & Brodie, D. (2020). Protocol-driven daily optimisation of venovenous extracorporeal membrane oxygenation blood flows: an alternate paradigm? *Journal of Thoracic Disease*, 12(11), 6854.
- Keller, S. P. (2019). Management of peripheral venoarterial extracorporeal membrane oxygenation in cardiogenic shock. *Critical Care Medicine*, 47(9), 1235.
- Burkhoff, D., Sayer, G., Doshi, D., & Uriel, N. (2015). Hemodynamics of mechanical circulatory support. *Journal of the American College of Cardiology*, 66(23), 2663–2674.